

UNIT-1

Introduction

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Contents

- Data Science: Introduction to Data Science
- Digital Universe
- Sources of Data
- Information Commons
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- OSEMN Framework

Applications of Data Science

- Healthcare and disease prediction
- Recommendation systems (Netflix, Amazon)
- Fraud detection in banking
- Speech and image recognition
- Targeted advertising and smart systems

Digital Universe

Digital Universe = all digital data created worldwide.

Includes social media, IoT, cloud, videos, and apps.

Data grows rapidly every day.

Important for business, research, and society.

5 V's of Digital Universe

1. Volume – Huge amount of data
2. Variety – Different types of data
3. Velocity – Fast data generation
4. Veracity – Accuracy of data
5. Value – Useful insights from data

Uses of Digital Universe

- Predictive analytics and forecasting
- Personalized recommendations
- Fraud detection and cybersecurity
- Smart cities and healthcare
- Business intelligence and research

Sources of Data

❖ Structured Data

- **Databases:** Relational databases (e.g., MySQL, PostgreSQL) and NoSQL databases (e.g., MongoDB).
- **Spreadsheets:** CSV, Excel files.

❖ Unstructured Data

- **Text Files:** Plain text, JSON, XML.
- **Multimedia:** Images, audio, video files.

❖ Semi-Structured Data

- **Logs:** Server logs, application logs.
- **Web Data:** HTML pages, web scraping outputs, APIs and Streaming Data
- **APIs:** **RESTful APIs, SOAP APIs for accessing web services.**
- **Streaming:** Real-time data from sources like Kafka, MQTT.

Uses of Digital Universe

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TYPES OF DATA

The data in it will be of four types.

1. Unstructured data: Word, PDF, Text, images, audio and video
2. Semi Structured data: XML data.
3. Meta Data: Data about data
4. Structured data: Relational data

Information Commons

Shared environment for data access and collaboration.
Supports research, learning, and innovation.
Provides datasets, tools, and computing resources.
Encourages open science and teamwork.

Components of Information Commons

- Data repositories
- Cloud and computing resources
- Collaborative workspaces
- Software and visualization tools
- Knowledge repositories and training programs

Data Science Project Life Cycle

- Business Understanding
- Data Understanding
- Data Preparation
- Exploratory Data Analysis
- Modeling, Evaluation, Deployment

Business Understanding

Identify the business problem.

Define project goals clearly.

Understand customer requirements.

Set success criteria for the project.

Data Understanding & Preparation

Collect and explore data.

Check structure and quality of data.

Handle missing values and outliers.

Clean and prepare data for analysis.

Exploratory Data Analysis & Modeling

- Analyze patterns and relationships.
- Use charts and graphs for insights.
- Select suitable ML algorithms.
- Train models using prepared data.

Model Evaluation & Deployment

- Test model performance.
- Check accuracy and reliability.
- Deploy model into real applications.
- Monitor and improve the model regularly.
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OSEMN Framework

O – Obtain data

S – Scrub/Clean data

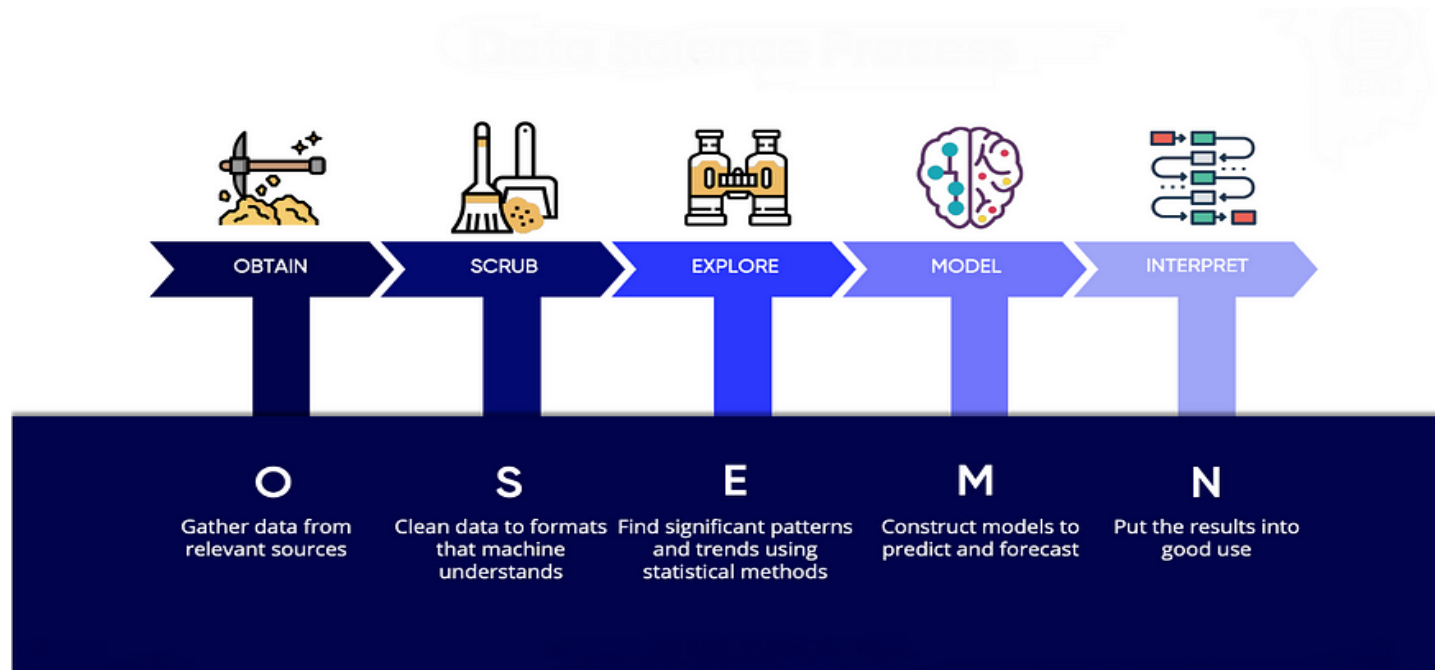
E – Explore data

M – Model the data

N – iNterpret and communicate results



OSEMN FRAMEWORK



References

1. Data Science for Business – Foster Provost and Tom Fawcett, *Data Science for Business*, O'Reilly Media, 2013.
2. Python for Data Analysis – Wes McKinney, *Python for Data Analysis*, 2nd Edition, O'Reilly Media, 2017.
3. Doing Data Science – Cathy O'Neil and Rachel Schutt, *Doing Data Science*, O'Reilly Media, 2013.
4. Data Science from Scratch – Joel Grus, *Data Science from Scratch*, 2nd Edition, O'Reilly Media, 2019.
5. Big Data Fundamentals – Thomas Erl, Wajid Khattak, and Paul Buhler, *Big Data Fundamentals*, Prentice Hall, 2015.
6. MEDIUM:"osemn framework", <https://medium.com>

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